

Lab # 03

Relational Operators in C++

Objective:

To understand and implement relational operators to compare values in C++.

Theory:

Relational operators are used to compare two values or variables. The result of a relational expression is either true (1) or false (0).

- ❖ Common Relational Operators
- ❖ Operator

Meaning:

- Equal to
==
- Not equal to
!=
- Greater than
>
- Less than
<
- Greater than or equal to
>=
- Less than or equal to
<=

Program:

```

#include <iostream>
using namespace std;

int main() {
    int a = 10, b = 5;
    // Do integer variables 'a' aur 'b' banaye gaye hain jinmein 10 aur 5 store hai

    cout << "a == b: " << (a == b) << endl;
    // Yeh check karta hai ke a aur b barabar hain ya nahi (equal)

    cout << "a != b: " << (a != b) << endl;
    // Yeh check karta hai ke a aur b barabar nahi hain (not equal)

    cout << "a > b: " << (a > b) << endl;
    // Yeh check karta hai ke a, b se bara hai ya nahi

    cout << "a < b: " << (a < b) << endl;
    // Yeh check karta hai ke a, b se chhota hai ya nahi

    cout << "a >= b: " << (a >= b) << endl;
    // Yeh check karta hai ke a, b se bara ya barabar hai

    cout << "a <= b: " << (a <= b) << endl;
    // Yeh check karta hai ke a, b se chhota ya barabar hai
    return 0;
}

```

Output:

```

a == b: 0
a != b: 1
a > b: 1
a < b: 0
a >= b: 1
a <= b: 0

```

Conclusion:

Relational operators help in comparing values and are commonly used in decision-making statements.

Logical Operators in C++

Title:

Study and Implementation of Logical Operators in C++

Objective:

To understand and use logical operators to combine conditions.

Theory:

Logical operators are used to combine multiple conditions and return a boolean result.

Types of Logical Operators:

❖ Operator

Meaning

- Logical AND

&&

- Logical NOT

!

Program:

```
#include <iostream>
using namespace std;

int main() {
    int a = 10, b = 5;
    // Do integer variables 'a' aur 'b' banaye gaye hain jinmein 10 aur 5 store hai

    cout << "Logical AND (a > 5 && b < 10): " << ((a > 5) && (b < 10)) << endl;
    // Yeh check karta hai ke dono conditions true hain ya nahi (AND operator)
    // a > 5 bhi true hai aur b < 10 bhi true hai to result true (1) hoga

    cout << "Logical OR (a < 5 || b < 10): " << ((a < 5) || (b < 10)) << endl;
    // Yeh check karta hai ke kam az kam ek condition true ho (OR operator)
    // a < 5 false hai lekin b < 10 true hai to result true (1) hoga

    cout << "Logical NOT !(a > b): " << !(a > b) << endl;
    // Yeh NOT operator hai jo result ko ulta kar deta hai
    // a > b true hai lekin NOT (!) usay false (0) bana deta hai

    return 0;
    // Program yahan end hota hai aur 0 return karta hai
}
```

Output:

Logical AND: 1
Logical OR: 1
Logical NOT: 0

Conclusion:

Logical operators are useful for combining multiple conditions in programs, especially in decision-making.

Conditional Statements in C++**Title:**

Study and Implementation of Conditional Statements in C++

Objective:

To understand decision-making using conditional statements like if, if-else, and switch.

Theory:

Conditional statements allow a program to make decisions based on conditions.

Types of Conditionals:

- if
- else if
- if-else
- switch

Program 1:**if-else****Code:**

```

#include <iostream>
using namespace std;

int main() {
    int num;
    // Ek integer variable 'num' banaya gaya hai user se input lene ke liye
    cout << "Enter a number: ";
    // User se number enter karne ke liye message show kiya gaya hai
    cin >> num;
    // User ka input 'num' variable mein store kiya jata hai
    if (num % 2 == 0) {
        // Yeh check karta hai ke number 2 se divide hone par remainder 0 hai ya nahi
        // Agar remainder 0 ho to number even hota hai
        cout << "Even number";
        // Agar condition true ho to "Even number" print hoga
    } else {
        // Agar condition false ho (remainder 0 na ho) to yeh block chalega
        cout << "Odd number";
        // Yeh "Odd number" print karega
    }
    return 0;
    // Program yahan end hota hai aur 0 return karta hai
}

```

Program 2:

else-if

Code:

```

#include <iostream>
using namespace std;

int main() {
    int marks;
    // Ek integer variable 'marks' banaya gaya hai user se marks lene ke liye
    cout << "Enter marks: ";
    // User ko message diya jata hai ke apne marks enter kare
    cin >> marks;
    // User ke enter kiye gaye marks 'marks' variable mein store hote hain
    if (marks >= 80) {
        // Yeh check karta hai ke marks 80 ya us se zyada hain
        cout << "Grade A";
        // Agar condition true ho to "Grade A" print hoga
    } else if (marks >= 60) {
        // Agar pehli condition false ho to yeh check hota hai ke marks 60 ya us se zyada hain
        cout << "Grade B";
        // Agar yeh condition true ho to "Grade B" print hoga
    } else if (marks >= 40) {
        // Agar upar wali conditions false ho to yeh check hota hai ke marks 40 ya us se zyada hain
        cout << "Grade C";
        // Agar yeh condition true ho to "Grade C" print hoga
    } else {
        // Agar koi bhi condition true na ho to yeh block chalega
        cout << "Fail";
        // Yeh "Fail" print karega
    }
    return 0;
}

```

Program 3:

switch statement

Code:

```
#include <iostream>
using namespace std;

int main() {
    int day;
    cout << "Enter day number (1-7): ";
    // User ko message diya jata hai ke 1 se 7 tak koi number enter kare
    cin >> day;
    // User ka input 'day' variable mein store ho jata hai
    switch(day) {
        // Switch statement use hota hai multiple conditions check karne ke liye
        case 1: cout << "Monday"; break;
        // Agar day 1 ho to Monday print hoga
        case 2: cout << "Tuesday"; break;
        // Agar day 2 ho to Tuesday print hoga
        case 3: cout << "Wednesday"; break;
        // Agar day 3 ho to Wednesday print hoga
        case 4: cout << "Thursday"; break;
        // Agar day 4 ho to Thursday print hoga
        case 5: cout << "Friday"; break;
        // Agar day 5 ho to Friday print hoga
        case 6: cout << "Saturday"; break;
        // Agar day 6 ho to Saturday print hoga
        case 7: cout << "Sunday"; break;
        // Agar day 7 ho to Sunday print hoga
        default: cout << "Invalid input";
        // Agar koi bhi case match na ho to "Invalid input" print hoga
    }
    return 0;}
```

Conclusion:

Conditional statements are essential for controlling the flow of a program and making decisions based on user input or conditions.

Tasks:

(1):

```
#include <iostream>
using namespace std;
int main()
{
    int num1, num2;
    cout << "Enter first number: ";
    // User se pehla number enter karne ko kaha jata hai
    cin >> num1;
    // Pehla number num1 variable mein store hota hai
    cout << "Enter second number: ";
    // User se doosra number enter karne ko kaha jata hai
    cin >> num2;
    // Doosra number num2 variable mein store hota hai
    // Ab dono numbers ka comparison kiya jata hai
    if (num1 == num2){
        // Agar dono numbers barabar hon
        cout << "Both numbers are equal.";
        // Message show hota hai ke dono numbers equal hain
    }
    else if (num1 > num2) {
        // Agar pehla number doosre se bada ho
        cout << "First number is greater than second number.";
        // Pehla number zyada hai
    }
    else {
        // Agar pehla number chhota ho doosre se
        cout << "First number is less than second number.";
    }
    return 0;}
```

(2):

```
#include <iostream>
using namespace std;
int main()
{
    int number;
    // Ek integer variable 'number' banaya gaya hai
    cout << "Enter an integer: ";
    // User se ek integer number enter karne ko kaha jata hai
    cin >> number;
    // User ka input 'number' variable mein store hota hai
    // Check kiya jata hai ke number 0 se bara aur 100 se chhota hai ya nahi
    if (number > 0 && number < 100)
    {
        // Agar dono conditions true hon (AND operator)
        cout << "The number is greater than 0 and less than 100.";
        // Message show hota hai ke number range ke andar hai
    }
    else
    {
        // Agar koi bhi condition false ho
        cout << "The number does NOT meet both conditions.";
        // Message show hota hai ke number range ke andar nahi hai
    }
    return 0;
    // Program yahan khatam hota hai
}
```

(3):

```
#include <iostream>
using namespace std;

int main()
{
    int num;
    // Ek integer variable 'num' banaya gaya hai
    cout << "Enter an integer: ";
    // User ko message diya jata hai ke ek number enter kare
    cin >> num;
    // User ka input 'num' variable mein store hota hai
    // Check kiya jata hai ke number positive hai ya nahi
    if (num > 0)
    {
        // Agar number 0 se bara ho

        cout << "The number is positive." << endl;
        // Message show hota hai ke number positive hai
    }
    else
    {
        // Agar number 0 ya negative ho

        cout << "The number is not positive." << endl;
        // Message show hota hai ke number positive nahi hai
    }
    return 0;
    // Program yahan khatam hota hai
}
```

(4):

```
#include <iostream>
using namespace std;
int main() {
    int age;
    // Ek integer variable 'age' banaya gaya hai
    cout << "Apni age enter karein: ";
    // User se age enter karne ko kaha jata hai
    cin >> age;
    // User ki di hui age 'age' variable mein store hoti hai

    if (age >= 18) {
        // Check kiya jata hai ke age 18 ya us se zyada hai ya nahi
        cout << "Aap adult hain." << endl;
        // Agar age 18 ya zyada ho to message show hota hai ke user adult hai
    }
    else {
        // Agar age 18 se kam ho
        cout << "Aap adult nahi hain." << endl;
        // Message show hota hai ke user adult nahi hai
    }
    return 0;
    // Program yahan khatam hota hai
}
```


(5):

```
#include <iostream>
using namespace std;
int main() {
    int number;
    // Ek integer variable 'number' banaya gaya hai
    cout << "Ek number enter karein: ";
    // User se ek number enter karne ko kaha jata hai
    cin >> number;
    // User ka input 'number' variable mein store hota hai
    if (number > 0) {
        // Check kiya jata hai ke number 0 se bara hai ya nahi
        cout << "Number positive hai." << endl;
        // Agar number 0 se bara ho to positive print hota hai
    }
    else if (number < 0) {
        // Check kiya jata hai ke number 0 se chota hai ya nahi

        cout << "Number negative hai." << endl;
        // Agar number 0 se chota ho to negative print hota hai
    }
    else {
        // Agar number na positive ho na negative (yaani 0 ho)

        cout << "Number zero hai." << endl;
        // Message show hota hai ke number zero hai
    }
    return 0;
    // Program yahan khatam hota hai
}
```
